

Polishing SOP

Steps 1-6 are copied word for word from [Ronnie's Google Doc](#) on 2026-10-05. Steps 7-11 are a draft that replaces the Doc's Step 7 ("Use a vibrating polisher").

Step 1

- Go to the Project Support Center and check out a diamond-edged saw blade. They should give you a bag with a couple blades in a cardboard container, a couple wrenches, and some metal pieces.
- Go back to the polishing lab to the saw machine. Lift up the platform to connect the saw blade to the machine.
- Locate the bolt directly under and to the right of the slit on the platform where the saw blade goes. Put the widest ring on the bolt, then the blade, then the other wide ring. The flat side of the wide rings should be touching the blade. Find the piece that fits into the divet of the wide ring, then screw the nut into the end of the bolt to hold everything in place. Tighten using the two wrenches.
- Make sure the water valve in the back is closed, and should be parallel to the back of the machine.
- Fill up the water bucket with water from the fume hood and fill up the machine until about 1/4-1/2" of the blade is sitting in the water.
- Lower the table back down.
- Place the sample against the two stabilizing rods, then clamp in place. Make sure the blade cover is down and the saw is not touching the sample. Make sure clamps are not in the way of the blade.
- Turn the machine on using the switch on the right side towards the back, then press the green button on the side, then the green button in front.
- Get RPM's to around 1400. The sample should be about 1cm^3 , just a size where it can fit on an SEM stub and won't spill over the edges of the stub. Slowly rotate the handle to move the saw, slowly cutting the sample and keeping RPMs around 1000-1100. Listen to make sure there are no weird noises. A little bit of water should be spraying up. If it is not, then check to make sure the blade is touching the water, since not having water could cause the blade to melt.
- Disassemble saw, open water valve, dry off the machine and saw pieces. Return bag to PSC.

Step 2

- Turn on the Bakelite compressor with a switch in the back and open the water valve to the left of the machine.
- Place a sample with the face you want to polish facing downwards. Lower the chamber using the arrows on the screen.
- Fill the chamber with 3/4 scoop of Bakelite.

- Place the cap piece on top with metal facing down and green marble facing up and flush with the top of the machine. It should be an airtight fit, so it will take some wiggling around to get in completely. You may need to hit it lightly with a hammer or wipe off any excess Bakelite from the cap or chamber to get it in.
- Cover the chamber with the lock bar.
- Press operate. Configure settings according to the laminated sheet hanging above the machine. Press play.
- Once complete, remove the lock bar, use the up arrow to push the piece out of the chamber, and pull it off the cap. It should come off pretty easily. Wipe off excess Bakelite from the cap and inside of the chamber.
- Turn off the Bakelite oven.

Step 3

- Turn on the sanding belt and open the water valve on top to get water flowing down the belt.
- Sand the puck by moving it side to side across the belt until the sample is completely exposed and all Bakelite has been removed from its surface. Be careful with fingers.
- Turn off the sander and shut the water valve on top of the sander as well as the big, yellow-handled water valve to the left of the machine.

Step 4

- Initial polisher testing: Turn on the machine and open water valve. Gentle settings (from top down) are 35, 30, 150. Set the clock to 30 seconds. Press the "link" button in the top right corner well and "H2O" button simultaneously to turn on the water, then the "link" and "play" buttons simultaneously to start it spinning. If nothing is wrong, press the stop button.
- Rinse off all parts of the puck, the metal puck holder, metal sand paper ring, and table the sandpaper sits on to get all previous particles off.
- **Wear gloves when handling the sample, do not contaminate the surface.**
- Press "link" and "up" arrow simultaneously to pull up the puck holder.
- Take off the splash guard ring and metal sand paper holding ring, then place 320-grit sandpaper. Replace the two rings. Place the puck in the metal holder with the sample on the sand paper.
- Turn on the water and start the sander as described above. Sand while continuously checking the surface, moving on to the next grit only when all scratches are the same size. Water should be positioned so the water is washing the sanded-off particles away from the sample. If not, the particles could scratch the sample. The timer can be set for a certain amount of time.
- Go through grits 320, 400, 600, 800, 1200 (verify these values, they could be off). Between every grit change, use the water to thoroughly clean gloves, the puck, the metal puck holder, metal sand paper ring, and table the sandpaper sits on to get all previous particles off. There is a 1200 plain and 1200 fine, the only difference is texture, would be worth

checking with claude to see if we actually need to do the 1200 fine. The guy helping us said the 1200 plain is usually sufficient.

Step 5

Switch out sandpaper for a polishing pad. Polishing pads can be used multiple times if they are used for the exact same alloy, be careful as not following this rule can cause cross-contamination. Peel off the back to expose the sticky side of the pad, making sure to keep the back part so you can re-attach it to the sticky side and store it later. Cut a small square of electroplated tape to form a tab that can be used to remove the polishing pad by sticking it on the back and wrapping around the side up to the top. Then place it on the table.

- Make sure the puck is completely dry using compressed air from the fume hood, and wipe off excess water from the sides and bottom of the puck using a paper towel. **DO NOT WIPE THE SAMPLE WITH THE PAPER TOWEL!** The puck can only touch cleaned surfaces from now on to prevent foreign particles from entering the polishing space and scratching it. Before starting, get a clean paper towel to place the puck onto after polishing, as well as a deep glass container and squirt bottle full of deionized water.
- Once the pad is in place and the puck is clean, move the puck holder over the top of the polishing pad. Use 1micron alumina suspension (Allide high-tech products, de-agglomerated) for the first polishing step. Apply 6-7 generous circles of alumina, focusing on the center and moving outwards since the alumina will slowly move to the outside of the pad as it spins.
- Use link+down to lower the puck onto the polishing pad. Set time to 3 minutes and press start. Don't use water on the polishing pad. Regularly apply alumina to the pad to keep well lubricated. You will be able to see the alumina moving towards the edges of the pad. The alumina is what is doing the polishing at this point.
- Every couple minutes, remove from the puck holder and wash with DI water, letting excess water drain into the deep glass container, checking for any scratches, then drying with compressed air. Repeat until there are no more visible scratches (verify this piece of info). Never set the polished side facing down.
- Wash alumina off of the polishing pad in the fume hood, as well as all of the polishing machine. Dry off the pad with compressed air before putting it in the bag. Put the sticky-side cover on the sticky side of the pad, then place in a bag labeled with the polishing material (such as suspended alumina) and what metal was polished on it.
- NOTES: All sand paper pads are single use, and if a pad starts to rip while polishing, replace it. The machine has to be reserved, there's a QR code to get to the website.

Step 6

- Take the puck, polished side up, to the fume hood in the deep glass container. Spray with ethanol to clean off the surface and dry with compressed air. Clean out the glass container twice by rinsing it out with ethanol and dumping it into the chemical waste. Now, the glass container and puck are both considered "clean".

- Place the puck faceup in the glass container and submerge the puck completely in methanol so the polished surface is not exposed to air.
- Clean out the ultrasonicator with deionized water, then fill with 1/4-1/2 inch of deionized water.
- Place the glass container into the water with a lid on top so no water gets into the glass container or touches the sample. Put the lid on top of the ultrasonicator and turn it on, cleaning for however long is needed.
- Bring it back to the fume hood. Remove the lid carefully so no water drips into methanol. Dry with compressed air. There may be a haze over the sample, and if there is, repeat ultrasonication until haze is no longer present.

Steps 7-11 are a draft. They were written from the #110 discussion of [09-28](#), [09-30](#) and [10-01](#), and nobody has run them yet, so expect to change them after the first sample. For EDS samples they replace the vibrating polisher. EBSD samples still need the colloidal silica on the vibrating polisher, which isn't written up yet.

Step 7

Final polish on 0.1 μm diamond film. EDS doesn't need the 5 hour colloidal silica polish (that is for EBSD), but aluminum still needs about a 0.1 μm finish for accurate EDS. The film leaves no liquid to dry on the face, and any diamond left behind is carbon, which we don't measure.

- The puck has to go through all of Step 6, ultrasonic included, before it goes on the film. About 1 minute in the ultrasonicator is enough, in methanol as in Step 6 or in IPA. One grain of 1 μm alumina carried onto the 0.1 μm film can scratch every sample polished on that film afterwards.
- The polishing pad should already be off and bagged (end of Step 5). Rinse every part the alumina touched, the same way as between grits in Step 4: the metal puck holder, the rings and the table. Then dry the table completely with compressed air. A grain left on the table shows through the film as a bump.
- Peel the backing off one 0.1 μm diamond film disc (PACE PDAA-01P12, 3 per pack), keeping the backing so you can put it back on for storage, like the pads. Add a tab of electroplated tape as in Step 5 so it can be pulled off later. Lay it on the clean, dry table and press it down firmly from the center outward so no air bubbles are trapped under it.
- Settings are lighter and slower than for the papers. PACE recommends 60-120 rpm for fine film and 2-5 lb (9-22 N) per sample, with the head and table turning the same direction. If the 150 in Step 4's gentle settings is the table speed, turn it down to about 100. Use the lowest force the machine allows, since too much force tears the film.
- Unlike the alumina pad, the film runs wet. Turn the water on (link + H₂O, as in Step 4) before lowering the puck and leave it on the whole time, positioned so it washes away from the sample. The water carries off what the film cuts; without it the film clogs and can tear.
- Use link+down to lower the puck onto the film, set the time to 1 minute and press start. After each run, remove the puck from the holder, rinse it with DI water into the deep glass

container, dry it with compressed air, and check it under the optical microscope. Never set the polished side facing down.

- Stop when the scratches from the alumina are gone. The film's own scratches are about 0.1 μm , mostly too fine to see in an optical microscope, so any scratch you can still see is left over from the alumina or from a stray grain. Expect a few minutes in total: PACE's rule of thumb is 30-90 seconds per micron removed, and the finest film cuts slowest. If alumina scratches are still there after about 10 minutes, the film may be worn, or the jump from 1 μm may be too big for it. The 0.5 μm film (PACE PDAA-05P12) between the alumina and this film would fix the second.
- Right after the last run, rinse the face with DI water before the machine's water can dry on it. If the machine runs on tap water, it dries to calcium and magnesium spots, and Mg is one of the elements EDS is measuring.
- Rinse the film, dry it with compressed air, put its backing back on, and store it in a bag labeled "0.1 μm diamond film" and the metal that was polished on it. Like the pads, use each film for one alloy only. Reuse it until it stops cutting (each run takes longer) or shows bald or worn patches.

Step 8

Clean the puck after the film.

- Clean as in Step 6: the puck face up in a clean glass container, covered in IPA or methanol, lid on, in the ultrasonicator for about 1 minute. The IPA or methanol always goes in the lidded glass container sitting in the water, never straight into the ultrasonicator tank, because both are flammable.
- Dry it completely with compressed air, including the edges of the face. The tape in Step 9 won't seal on a wet surface, and it would trap whatever is under it.

Step 9

Tape the face and cut the sample out of the puck. The tape keeps the saw's water and grit off the polished face. What matters is that its edges stay sealed: in May the tape lifted at the edges during the cut and left a film along them ([#77](#)).

- Cover the whole face with Kapton tape: the metal plus 2-3 mm of Bakelite on every side. The roll in the cabinet is 1/4" (6 mm) wide, narrower than a ~1 cm sample, so lay strips side by side in the same direction, each overlapping the one before by a couple of millimeters.
- Press every edge and seam down firmly with a clean gloved finger, working out any bubbles. The edges and seams are where water gets in.
- Set up the diamond saw as in Step 1. Clamp the puck by the Bakelite, never across the taped face.
- Cut through the Bakelite 1-2 mm outside the metal on all four sides. That way, if the tape lifts at a cut edge, it lifts over Bakelite instead of over the polished metal.

- Then cut parallel to the face, a few millimeters less deep than the sample is tall, so the cut goes through the metal. The piece then has a bare metal back, which is what conducts to the SEM stub (Bakelite doesn't conduct), and it is no taller than about 1 cm, as in Step 1.
- Do the four side cuts first, while there's plenty of Bakelite to clamp, and the parallel cut last. Slow down at the end of each cut. If the piece drops into the water, only reach for it after the blade has stopped.
- With the tape still on, rinse the piece with DI water and dry it, so saw slurry on the sides can't run onto the face when the tape comes off.
- Peel the tape off slowly, starting from a Bakelite corner. If the tape stayed down, just blow the face off with compressed air; it doesn't need another solvent clean. If an edge lifted and left a film or stain, cleaning won't rescue that strip, so keep EDS away from it.
- Disassemble the saw and return the bag to the PSC, as at the end of Step 1.

Step 10

Mount the sample on an SEM stub.

- Wear gloves, and handle the piece only by its sides with clean tweezers.
- Stick a 12 mm carbon tab (Ted Pella 16084-1) on a clean pin stub and peel off the tab's top liner.
- Set the piece on the tab with the bare metal back down and the polished face up, centered so nothing hangs over the edge of the stub. Press it down by its sides, never on the face.
- If Bakelite ended up between the metal and the tab, the sample isn't grounded. Run a strip of carbon tape, or a thin line of the silver paint, from the edge of the metal over the Bakelite and down to the stub, well away from where you'll take EDS. Let silver paint dry completely before the stub goes into the vacuum jar or the SEM.
- Write down which stub is which (sample, alloy, date, and "0.1 μm diamond film"), for example by its position in the vacuum jar.

Step 11

Store the stub under vacuum until the SEM ([caliber#16](#)).

- Stand the stub by its pin in one of the holes in the vacuum jar's blue stub plate. Keep the desiccant pack from touching any stub.
- Close the lid, connect the pump hose to the jar, open the jar's valve (the black knob), and run the pump for about 45 seconds.
- Close the jar's valve before you switch the pump off. An oil-sealed pump that stops while it is still open to the jar can draw oil back up the hose toward the samples.
- Wait a few minutes and check that the lid is held down before putting the pump away. The seal has failed in the first few minutes once before.
- To open the jar, turn the valve slowly so the air comes back in gently. A rush of air blows dust around and can knock stubs over.
- Re-pump about once a month. The seal was still holding after 16 days.

- If the O-ring ever needs grease, use a thin film of Apiezon (hydrocarbon) grease, never silicone grease. Silicone shows up as Si in EDS.

Checking the first sample

- At the SEM, take SE and BSE images at 50,000x or more before trusting any numbers. Look for leftover scratches and for dark specks in BSE, which would be embedded diamond. Run EDS at 5 kV with C and O left in the element list.
- If you can, put the pure Al pellet through Steps 7-10 with it. Pure Al has no Si, so any Si on the pellet came from the process.

Open questions

- Which of Step 4's gentle settings (35, 30, 150) is the force, the head speed and the table speed. Step 7 needs the table at about 100 rpm and the lowest force.
- Does the SEM have a holder for 1 1/4" mounts? If it does, Steps 9 and 10 can be skipped: a strip of carbon tape from the edge of the metal face to the holder grounds the sample. Worth asking Mike.
- The Kapton roll is 1/4" wide. A 1" roll would cover the face in one piece, with no seams.
- The ASTM E1078 notes in [#77](#) say not to spray the specimen with compressed air, and this SOP uses the fume hood's compressed air throughout. Worth asking Mike whether that air is filtered, or whether to use nitrogen for the last dry in Step 8.

Sources for Steps 7-11

- Film settings, timing and replacement: PACE Technologies, [Lapping Films](#).
- Aluminum needs a 0.1 μm finish: Geller & Engle, NIST, [doi:10.6028/jres.107.051](#) ("Aluminum and copper: these metals scratch easily, requiring 0.1 μm to finish").
- Why diamond film and not SiC or silica: the [09-28 write-up](#).